

Program 2 : Develop a multi stage docker file for container orchestration.

File structure:

/package.json

/package-lock.json

/src/index.js

/node_modules

/Dockerfile

Steps to create project files:

1. Initialize Node.js project and install dependencies:

```
$ npm init -y
```

```
$ npm install express
```

2. In package.json, add the following in the "scripts" section:

```
"scripts": {  
  "start": "node dist/index.js",  
  "build": "mkdir -p dist && cp -r src/* dist/",  
  "test": "echo \"Error: no test specified\" && exit 1"  
}
```

3. Create Dockerfile and add the following commands:

```
-----  
FROM node:20-alpine AS builder  
LABEL author="Girish Yandigeri"  
WORKDIR /app  
COPY package.json package-lock.json ./  
RUN npm install  
COPY . .  
RUN npm run build  
  
FROM node:20-alpine  
WORKDIR /app  
COPY --from=builder /app/package.json ./
```

```
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules
EXPOSE 3000
CMD ["node", "dist/index.js"]
```

To create the image:

```
$ docker build -t program2 .
```

To list images:

```
$ docker images
```

To run container:

```
$ docker run -d -p 3000:3000 --name program2 program2
```

To see running containers:

```
$ docker ps
```

To test application:

```
$ curl http://localhost:3000
```

To check the stages used in build process:

```
$ docker history program2
```

Execution Output:

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab2$ ls
```

```
Dockerfile  node_modules  package.json  package-lock.json  src
```

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab2$ docker build -t program2 .
```

```
[+] Building 48.4s (15/15) FINISHED
```

```
docker:default
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 417B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> => resolve docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> => sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435 7.67kB / 7.67kB
=> => sha256:2b56f2779663b9e1a77bdb5235dc31f1a81e534ccab1c1b35c716a8db79eeab9 6.42kB / 6.42kB
=> => sha256:bebd32d651b3e0c9c2b28fcd1d3888408125d703232013cfff955344d052027e5 1.72kB / 1.72kB
=> => sha256:2d35ebdb57d9971fea0cac1582aa78935adff8058b2cc32db163c98822e5dfalb 3.80MB / 3.80MB
=> => sha256:60e45a9660cfaebbbac9bba98180aa28b3966b7f2462d132c46f51a1f5b25a64 42.75MB / 42.75MB
=> => sha256:e74e4ed823e9560b3fe51c0cab47dbfdcf4b12453604319408ec58708fb9e720 1.26MB / 1.26MB
=> => extracting sha256:2d35ebdb57d9971fea0cac1582aa78935adff8058b2cc32db163c98822e5dfalb
=> => sha256:da04d522c98fe12816b2bcdcf8413fca73645f8fa60f287c672f58bcc7f0fa38 444B / 444B
=> => extracting sha256:60e45a9660cfaebbbac9bba98180aa28b3966b7f2462d132c46f51a1f5b25a64
=> => extracting sha256:e74e4ed823e9560b3fe51c0cab47dbfdcf4b12453604319408ec58708fb9e720
=> => extracting sha256:da04d522c98fe12816b2bcdcf8413fca73645f8fa60f287c672f58bcc7f0fa38
=> [internal] load build context
=> => transferring context: 2.70MB
=> [builder 2/6] WORKDIR /app
=> [builder 3/6] COPY package.json package-lock.json ./
=> [builder 4/6] RUN npm install
=> [builder 5/6] COPY ...
=> [builder 6/6] RUN npm run build
=> [stage 1 3/6] COPY --from=builder /app/package.json ./
=> [stage 1 4/6] COPY --from=builder /app/package-lock.json ./
=> [stage 1 5/6] COPY --from=builder /app/dist ./dist
=> [stage 1 6/6] COPY --from=builder /app/node_modules ./node_modules
=> exporting to image
=> => exporting layers
=> => writing image sha256:de2177f6d4493bd101f3e8137ec47681ba0952729bd0500d92ee14517d1a89d6
=> => naming to docker.io/library/program2
```

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab2$ docker run -d -p 3000:3000 program2
```

```
c83b2fdaf18c395d72e54e09f179a90f85fd72652344aa502a09609ba5f77366
```

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab2$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
c83b2fdaf18c	program2	"docker-entrypoint.s..."	8 seconds ago	Up 6 seconds	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	gallant_galois
38c67a115ddf	program1	"docker-entrypoint.s..."	9 minutes ago	Up 9 minutes	0.0.0.0:4000->4000/tcp, [::]:4000->4000/tcp, 5000/tcp	mystifying_tharp

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab2$ curl http://localhost:3000
```

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab2$
```